

Econometrics I

Preliminary & Auxiliary Material: The Multivariate Normal Distribution

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Purpose of this material

- These notes are meant to be used as preliminary and/or auxiliary material for lecture 6.
- The goal of lecture 6 is to derive:
 - ① The asymptotic distribution of $\hat{\beta}$.
 - ② An estimator of the asymptotic variance of $\hat{\beta}$.
 - ③ The distributions of test statistics for common hypothesis tests.
 - ④ Asymptotic confidence intervals.
- In order to be able to do all of the above, we will need some results on the multivariate normal distribution, or **multinormal distribution**.
- In these notes, I will only state the results, without proofs.
 - We will only need to understand the results and how to use them.
 - I will not present the proofs because they require quite a bit of background in linear algebra in order to understand them, and thus, they are beyond the scope of this course.
- However, if you are interested in the details, you can check chapter 18 of Goldberger.
- Notation: **bold letters**¹ represent column vectors or matrices, like $\mathbf{x}$ and $\mathbf{M}$.

¹In my blackboard lecture notes, I represent vectors by underlining them because it is difficult to recognize bold letters in handwriting.

- Let $\mathbf{y}$ be a $n \times 1$ random vector, whose joint pdf is given by

$$f(\mathbf{y}) = (2\pi)^{-n/2} |\Sigma|^{-1/2} e^{-w/2}$$

where

- $w = (\mathbf{y} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{y} - \boldsymbol{\mu})$ is a scalar.
- $\boldsymbol{\mu}$ is a $n \times 1$ parameter vector to be interpreted as a vector of means.
- Σ is a $n \times n$ positive definite parameter matrix to be interpreted as the variance-covariance matrix.
- $|\Sigma|^{-1/2} = 1/\sqrt{\det(\Sigma)}$ where $\det$ denotes the determinant.
- Then we say that the distribution of $\mathbf{y}$ is multivariate normal or multinormal and $\mathbf{y} \sim N(\boldsymbol{\mu}, \Sigma)$
- Remark: notice that the scalar w is actually a **quadratic form**.

Two special cases

- Consider the case $n = 1$, then $\mathbf{y} = y$, $\boldsymbol{\mu} = \mu$ and $\boldsymbol{\Sigma} = \sigma^2$, $w = z^2$, with $z = (y - \mu)/\sigma$. So $f(y)$ is the familiar normal density: $f(y) = \exp(-z^2/2)/\sqrt{2\pi\sigma^2}$.
- Consider the case $n = 2$ then, $\mathbf{y} = (y_1, y_2)^\top$, $\boldsymbol{\mu} = (\mu_1, \mu_2)$, $\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$.
- Thus $\det(\boldsymbol{\Sigma}) = \sigma_{11}\sigma_{22} - \sigma_{12}^2 = \sigma_1^2\sigma_2^2(1 - \rho^2)$ where $\rho = \sigma_{12}/(\sigma_1\sigma_2)$, $\sigma_1^2 = \sigma_{11}$ and $\sigma_2^2 = \sigma_{22}$ and $\boldsymbol{\Sigma}^{-1} = (1 - \rho^2)^{-1} \begin{bmatrix} 1/\sigma_1^2 & -\rho/(\sigma_1\sigma_2) \\ -\rho/(\sigma_1\sigma_2) & 1/\sigma_2^2 \end{bmatrix}$
- Also $w = (z_1^2 + z_2^2 - 2\rho z_1 z_2)/(1 - \rho^2)$, with $z_1 = (y - \mu_1)/\sigma_1$, $z_2 = (y - \mu_2)/\sigma_2$.
- Some remarks are in order: σ_1^2 and σ_2^2 are the variances of y_1 and y_2 , σ_{12} is the covariance between y_1 and y_2 , and ρ is the correlation coefficient. In this case $f(y_1, y_2)$ follows a standard parameterization of the bivariate normal, in terms of its means, standard deviations and correlation coefficient.

Properties of a multinormal vector

- Return to the general case, partition the $n \times 1$ vector $\mathbf{y}$ into a $n_1 \times 1$ vector $\mathbf{y}_1$ and a $n_2 \times 1$ vector $\mathbf{y}_2$.
- Do the same for $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$:

$$\mathbf{y} = \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix}$$

- The following is a restatement, using matrix language, of properties we had reviewed before. If $\mathbf{y} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then:

P1: expectation and variance

The expectation and variance of $\mathbf{y}$ are respectively: $E(\mathbf{y}) = \boldsymbol{\mu}$ and $V(\mathbf{y}) = \boldsymbol{\Sigma}$.

P2: Normal Marginals

The marginal distribution of $\mathbf{y}_1$ is multinormal: $\mathbf{y}_1 \sim N(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$

P3: Conditional distribution is multinormal

The conditional distribution of $\mathbf{y}_2$ given $\mathbf{y}_1$ is multinormal: $\mathbf{y}_2|\mathbf{y}_1 \sim N(\boldsymbol{\mu}_2^*, \boldsymbol{\Sigma}_{22}^*)$ where

$$\boldsymbol{\mu}_2^* = E(\mathbf{y}_2|\mathbf{y}_1) = \boldsymbol{\alpha} + \mathbf{B}^\top \mathbf{y}_1,$$

$$\mathbf{B} = \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\Sigma}_{12},$$

$$\boldsymbol{\alpha} = \boldsymbol{\mu}_2 - \mathbf{B}^\top \boldsymbol{\mu}_1,$$

$$\boldsymbol{\Sigma}_{22}^* = V(\mathbf{y}_2|\mathbf{y}_1) = \boldsymbol{\Sigma}_{22} - \mathbf{B}^\top \boldsymbol{\Sigma}_{11} \mathbf{B}$$

Remarks: Note that the conditional expectation function $E(\mathbf{y}_2|\mathbf{y}_1)$ is linear and that the conditional variance-covariance matrix is constant across $\mathbf{y}_1$.

P4: Uncorrelatedness implies independence

If $\Sigma_{12} = \mathbf{0}$, then $\mathbf{y}_1$ and $\mathbf{y}_2$ are independent random vectors.

Note that the roles of $\mathbf{y}_1$ and $\mathbf{y}_2$ can be reversed

Proof.

I'm just presenting this proof because it is quite simple and immediate from the definitions. If $\Sigma_{12} = \mathbf{0}$ then $\mathbf{B} = \mathbf{0}$ so $\mu_2^* = \alpha = \mu_2$ and $\Sigma_{22}^* = \Sigma_{12}$. Thus $\mathbf{y}_2|\mathbf{y}_1 \sim N(\mu_2, \Sigma_{22})$, for all $\mathbf{y}_1$. These conditional distributions are all the same and they coincide with the marginal distribution. Thus the vectors are stochastically independent. □

Special case 1: $n_1 = n_2 = 1$

This is the bivariate normal distribution, P2 and P3 Specialize respectively to $y_1 \sim N(\mu_1, \sigma_{11})$, $y_2|y_1 \sim N(\alpha + \beta y_1, \sigma^2)$, where $\beta = \sigma_{12}/\sigma_{11}$, $\alpha = \mu_2 - \beta\mu_1$, $\sigma^2 = \sigma_{11} - \beta^2\sigma_{11}$.

Special case 2: $n_1 = n - 1$ and $n_2 = 1$

Here y_2 is a scalar and $\mathbf{y}_1$ is a vector. The variables and parameters given this partition are

$$\mathbf{y} = \begin{bmatrix} \mathbf{y}_1 \\ y_2 \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \mu_2 \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\sigma}_{12} \\ \boldsymbol{\sigma}_{12}^\top & \sigma_{22} \end{bmatrix}$$

Here we have that $y_2|\mathbf{y}_1 \sim N(\alpha + \boldsymbol{\beta}^\top \mathbf{y}_1, \sigma^2)$ where $\boldsymbol{\beta} = \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\sigma}_{12}$, $\alpha = \mu_2 - \boldsymbol{\beta}^\top \boldsymbol{\mu}_1$, $\sigma^2 = \sigma_{22} - \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{11} \boldsymbol{\beta}$. Notice that this case satisfies the assumptions of a **conditional approach for a multiple linear regression model**, i.e., the conditional expectation is linear $E(y_2|\mathbf{y}_1) = \alpha + \boldsymbol{\beta}^\top \mathbf{y}_1$ and the conditional variance $V(y_2|\mathbf{y}_1) = \sigma^2$ is constant.

P5: Linear functions of a multinormal vector are multinormal

If $\mathbf{z} = \mathbf{g} + \mathbf{H}\mathbf{y}$, where $\mathbf{g}$ and $\mathbf{H}$ are nonrandom, and $\mathbf{H}$ has full row rank, then $\mathbf{z} \sim N(\mathbf{g} + \mathbf{H}\boldsymbol{\mu}, \mathbf{H}\boldsymbol{\Sigma}\mathbf{H}^\top)$.

- Why $\mathbf{H}$ needs to have full row rank? Remember that the row rank of a matrix is the number of *linearly independent rows*. This condition is required to rule out degeneracies, in order to ensure that $V(\mathbf{z})$ is nonsingular, which is a prerequisite for multinormality.
- Note also that if $\mathbf{y}$ is $n \times 1$ and $\mathbf{H}$ is $m \times n$ with $m > n$ then $\mathbf{H}$ cannot have full row rank. Thus, in general, we work with matrices $\mathbf{H}$ with $m < n$ and such that the row rank is equal to m . If we interpret n as the number of observations in a random sample, as in the applications we deal with, we can expect to have $m < n$.

- If the $n \times 1$ vector $\mathbf{z}$ is multinormal with $N(\mathbf{0}, \mathbf{I})$, then we say that $\mathbf{z}$ is a standard normal vector.
- Then $z_1, \dots, z_n$ are pairwise independent standard normal random variables.
- The multinormal pdf simplifies to
 $f(\mathbf{z}) = (2\pi)^{-n/2} \exp(-\mathbf{z}'\mathbf{z}/2) = \prod_{i=1}^n \exp((-z_i^2/2)/\sqrt{2\pi})$ which is the product of n independent standard normal pdf's.

Chi-square distribution with n degrees of freedom $\chi^2(n)$

If $w = \mathbf{z}^T \mathbf{z}$ where $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I})$ then $w \sim \chi^2(n)$. This is, the sum squares of n independent standard normal random variables has the chi-square distribution with parameter n .

F distribution with m and n degrees of freedom $F(m, n)$

If $v = \frac{(w_1/m)}{(w_2/n)}$, where $w_1 \sim \chi^2(m)$ and $w_2 \sim \chi^2(n)$ are independent, then $v \sim F(m, n)$. That is, the ratio of two independent chi-square variables, each divided by its degrees of freedom, has the F distribution, with degrees of freedom equal to the respective chi-squares.

Student's t -distribution

If $u = \frac{z}{\sqrt{w/n}}$ where $z \sim N(0, 1)$ and $w \sim \chi^2(n)$ are independent, then $u \sim t(n)$. That is, the ratio of a standard normal random variable to the square root of an independent chi-square random variable divided by its degrees of freedom has the Student's t -distribution

Remark

Observe that if $u \sim t(n)$, then $u^2 \sim F(1, n)$.

Quadratic Forms in Normal Vectors

- Now we are ready to state the important results from this section.

Q1

Suppose that the $n \times 1$ vector $\mathbf{y} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Let $w = (\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})$. Then $w \sim \chi^2(n)$.

Q2

Suppose that the $n \times 1$ vector $\mathbf{u} \sim N(\mathbf{0}, \mathbf{I})$. Let $\mathbf{M}$ be a nonrandom $n \times n$ idempotent matrix with $\text{rank}(\mathbf{M}) = r \leq n$, i.e., $\mathbf{M}$ doesn't have full rank. Let $w = \mathbf{u}^\top \mathbf{M} \mathbf{u}$. Then $w = \mathbf{u}^\top \mathbf{M} \mathbf{u} \sim \chi^2(r)$.

Q3

Suppose that the $n \times 1$ vector $\mathbf{u} \sim N(\mathbf{0}, \mathbf{I})$. Let $\mathbf{M}$ be a nonrandom $n \times n$ idempotent matrix with $\text{rank}(\mathbf{M}) = r \leq n$. Let $\mathbf{L}$ be a nonrandom matrix such that $\mathbf{L}\mathbf{M} = \mathbf{O}$, where $\mathbf{O}$ is a matrix of only zeros. Let $\mathbf{t}_1 = \mathbf{M}\mathbf{u}$ and $\mathbf{t}_2 = \mathbf{L}\mathbf{u}$. Then $\mathbf{t}_1$ and $\mathbf{t}_2$ are independent random vectors.